

Mathematical appendix: inference, control, and the cost of learning

This appendix develops the exact benchmark likelihood and a task-specific finite-sample lower bound. The lower bound concerns the explicitly defined stationary reduction below, not the full switching-regime control problem. Numerical illustrations are generated by `verification/verify_theory.py` and accompany the derivation; simulation does not substitute for proof.

1. Observation law and filtering

Write $\mu = .03$, $\sigma = .30$, $d_k = h + k\Delta$, $\ell = h + c_T = .027$. For a submitted side $s \in \{-1, +1\}$ at depth k_s , let

$$a_{s,k_s}(x) = \Phi^{-1}\{g(-.3 - .7k_s - .2sx)\}, \quad z = \frac{r - \mu x}{\sigma}.$$

Transforming the genuinely observed return r to z uses only the public return model; it does not expose an innovation before the action. Given current regime $h_0 \in \{-1, +1\}$, define

$$c_{h_0,s}(z, x) = \Phi\left(\frac{a_{s,k_s}(x) - h_0\theta sz}{\sqrt{1 - \theta^2}}\right). \quad (1)$$

For action a with submitted-side set $S(a)$, the joint density of the return and selected fills is

$$L_{h_0}(r, f \mid x, a) = \frac{\phi(z)}{\sigma} \prod_{s \in S(a)} c_{h_0,s}(z, x)^{f_s} \{1 - c_{h_0,s}(z, x)\}^{1-f_s}. \quad (2)$$

The product is valid because side-specific primitive noises are independent **conditional on both z and h_0** . It is not a product of unconditional marginal fill probabilities. At prior $b = \Pr(H = +1 \mid \mathcal{I})$, the observable density is $(1 - b)L_- + bL_+$; the mixture is outside the product.

Let $\ell_\pm$ be the product in (2), without the common return density. The current-regime posterior and next-period decision-time belief are

$$B(b, z, f, a, x) = \frac{b\ell_+}{(1 - b)\ell_- + b\ell_+}, \quad b_{\text{next}} = \kappa + (1 - 2\kappa)B. \quad (3)$$

First infer H_t , then transition to H_{t+1} . The independent signal transition adds no regime information. Abstinence and market actions have an empty product: $\ell_+ = \ell_- = 1$, $B = b$, and only prediction toward $1/2$ remains. A submitted non-fill contributes $1 - c$; a missing side contributes nothing.

Use $\log \Phi(v)$ for a fill and $\log \Phi(-v)$ for a non-fill, sum over selected sides, and normalise the two weighted log likelihoods using log-sum-exp. Do not compute $1 - \Phi(v)$ in a far tail. Priors $b = 0, 1$ remain endpoints after measurement and become $\kappa, 1 - \kappa$ after prediction; handle their log weights explicitly.

1.1 What carries information?

Public returns and signals have laws independent of the sign regime. Each fill marginal is also sign-independent. More strongly, the **entire contemporaneous selected fill vector** has the same law under either sign:

$$\Pr_+(F = f \mid x, a) = \int \phi(z)\ell_+(z, f) dz = \int \phi(z)\ell_-(-z, f) dz = \Pr_-(F = f \mid x, a). \quad (4)$$

This follows from $\ell_+(z, f) = \ell_-(-z, f)$ and holds at asymmetric depths and $x \neq 0$. It does not imply independent unconditional fills. Both-side latent execution variables have covariance $-\theta^2$. Their joint fill law can reveal dependence magnitude when unknown, but not the current sign at a fixed magnitude.

Joint return/execution feedback reveals the sign through (1), including informative non-fills. The current fill vector, and thus post-execution inventory at a fixed state/action, carries no sign evidence alone. In particular, $\mathbb{E}[B \mid F = f, x, a] = b$.

Regime-classification accuracy, Brier scores against H_t , and detection delays measured from actual switches require privileged labels. Observed-data log scores, joint return/fill residuals, selected-fill calibration, quote coverage and evidence age can run online. Marginal fill or return checks can pass while dependence is wrong. Monitoring remains subject to endogenous selection: no quotes means no fresh execution evidence.

1.2 Unknown parameters on a declared finite support

Index a public candidate pair (θ_j, κ_j) by j . Let w_j be its posterior probability and b_j its conditional regime belief. Writing $m_j = (1 - b_j)\ell_{j,-} + b_j\ell_{j,+}$, the exact joint update is

$$w_j^+ = \frac{w_j m_j}{\sum_i w_i m_i}, \quad b_j^+ = \frac{b_j \ell_{j,+}}{m_j}, \quad b_{j,\text{next}} = \kappa_j + (1 - 2\kappa_j)b_j^+. \quad (5)$$

Independent episodes reset $b_j = 1/2$. Carry model weights across permitted pilot episodes; reset each test episode to its pilot-fitted weights. No-quote steps have $m_j = 1$ and leave weights unchanged. Given the public past, the same behavioural action kernel cancels from every model likelihood, so no importance weighting is needed. This fails for actions chosen using hidden labels.

The joint posterior $(w_j, b_j)_j$ is a sufficient inference state on the declared support. Averaging known-model Q values permits separately optimised future actions for each model: an optimistic relaxation, not joint Bayes-adaptive control. When pilots nearly identify the parameters, its empirical comparison principally tests future **regime** information; it does not establish the value of learning persistently unknown parameters.

2. Wealth identity and sufficient control state

Put $W_t = C_t + Q_t P_t$. For passive actions let $\Delta q = \sum_s s f_s$ and $G = \sum_s (d_{k_s} - c_P) f_s$. For a market action s , let $\Delta q = s$, $G = -\ell$. Both are zero for abstention. The signed cash ledger gives the pathwise identity

$$W_{t+1} - W_t = G_t + (Q_t + \Delta q_t)R_t = G_t + Q_{t+1}R_t. \quad (6)$$

Consequently

$$\Pi_T = \sum_{t < T} \{G_t + Q_{t+1}R_t\} - \ell|Q_T|, \quad J = \mathbb{E} \left[\Pi_T - \lambda \sum_{t < T} Q_t^2 \right]. \quad (7)$$

The inventory penalty does not debit cash. Both-side fills cancel inventory but retain both spread/fee contributions. Abstention preserves return exposure on existing inventory. Terminal liquidation follows the final return, is charged once, and can close multiple units.

Split G into passive spread revenue, passive fees, discretionary market spread and market fees. Split exposure returns into

$$Q_{t+1}R_t = \mu X_t Q_{t+1} + \sigma Z_t Q_t + \sigma Z_t \Delta q_t^{\text{market}} + \sigma Z_t \Delta q_t^{\text{passive}}. \quad (8)$$

These are signal-associated exposure, noise on inherited inventory, noise on predictable market trades, and execution selection. The middle two have zero conditional mean; passive selection generally does not. This is pathwise accounting, not a claim that realised directional attribution is risk-free.

Translation invariance removes absolute cash and price from control. All constraints depend on inventory; the return/fill laws depend on (x, H) ; future increments in (6) do not depend on C_t, P_t . No cash budget, price-dependent action constraint, or nonlinear wealth utility is imposed. The known-parameter sufficient state is (m, q, x, b) , where m is remaining decisions. These particular assumptions justify removing cash and midpoint.

2.1 Exact Bellman recursion

Let $V_m(q, x, b)$ be optimal future wealth increments, less utility penalties and terminal liquidation. Its boundary is

$$V_0(q, x, b) = -\ell|q|. \quad (9)$$

All maximisations use actions admissible under **every** possible fill subset. Put $\bar{\ell}_b = (1 - b)\ell_- + b\ell_+$, $\Psi(u) = \kappa + (1 - 2\kappa)u$, and $q' = q + \Delta q(a, f)$. Then

$$V_m(q, x, b) = \max_{a \in \mathcal{A}(q)} \left\{ -\lambda q^2 + \int \phi(z) \sum_f \bar{\ell}_b(z, f) \left[G(a, f) + q'(\mu x + \sigma z) + \sum_{x'} K_{xx'} V_{m-1}\{q', x', \Psi(B)\} \right] dz \right\}. \quad (10)$$

For market/abstention actions the sum is a singleton and the feedback likelihood is one. The initial objective is $V_T(0, 0, 1/2)$. Quadrature and discretisation approximate (10); refinement supports numerical stability, not a certified error bound.

Immediate expected reward has a closed form. With $V_s = H\theta sZ + \sqrt{1 - \theta^2}U_s$, joint normality gives $\mathbb{E}[Z \mid V_s] = H\theta sV_s$, so

$$\mathbb{E}[ZF_s \mid H, x, a] = -H\theta s\phi(a_{s,k_s}). \quad (11)$$

The passive expected stage increment including its penalty is

$$-\lambda q^2 + q\mu x + \sum_{s \in S(a)} \left[p_{s,k_s}(x)(d_{k_s} - c_P + s\mu x) - \sigma\theta(2b-1)\phi(a_{s,k_s}) \right]. \quad (12)$$

For a market action it is $-\lambda q^2 + (q+s)\mu x - \ell$. The mean sign $2b-1$ is valid in this **linear expectation**; replacing the nonlinear likelihood by a mean-correlation likelihood is invalid.

2.2 Full information and the no-information-value comparison

The full-information value $U_m(q, x, H)$ has identical boundary, actions, reward and constraints. Its observation kernel is ℓ_H , and continuation is

$$\sum_{x'} K_{xx'} \sum_{H'} T_\kappa(H, H') U_{m-1}(q', x', H'). \quad (13)$$

It sees current H , not future states or innovations. Its exact optimum weakly dominates the partially observed optimum under the same initial regime distribution.

For a no-learning planning recursion $\tilde{V}$, replace $\Psi(B)$ in (10) by $\Psi(b)$, retaining immediate economics and realised inventory transitions. Equation (4) makes this a coherent planning problem that ignores joint return/fill learning: the inventory transition itself gives no sign evidence.

Finite-horizon values are convex in b : each fixed history-dependent policy's expected reward is affine in the initial mixture, and its supremum is convex. For any convex continuation $v(q', x', \cdot)$, conditioning on the fill vector in (4) yields

$$\mathbb{E}[v(q', x', \Psi(B)) \mid F = f] \geq v(q', x', \Psi(b)). \quad (14)$$

This common-continuation comparison isolates the benefit of feedback-conditioned future choices. Induction gives $V_m \geq \tilde{V}_m$ for exact known-parameter planning values.

The matched baseline nevertheless filters actual feedback and replans at every real step. Its deployed return is not $\tilde{V}_T$, which supposes all future learning is ignored. Compare deployed returns directly: a gap between offline planning values alone does not establish a material advantage over the filtering baseline. Numerical approximations and unknown-model mixtures also preclude treating the practical controller as exactly Bayes-optimal.

3. A stationary repeated one-step reduction

Fix $\theta = .35$. Environment $\mathcal{E}_0$ has permanent $H = +1$; $\mathcal{E}_1$ has permanent $H = -1$. The unknown sign never switches. Each of n opportunities draws a fresh independent triple (Z, U_+, U_-) , observes fixed $X = 0$, starts at midpoint 1000, and **resets pre-decision inventory and cash to zero**. The public-signal chain and inter-period inventory accumulation are removed. Liquidation occurs after each opportunity's return with cost $\ell|q'|$. Pre-decision inventory penalties vanish.

Keep all eleven original actions: either depth on either side, both market directions and abstention. The original inventory bound five may remain; even a bound of one admits all actions at a reset state. Feedback is the public return and only selected fills. PnL is a deterministic function of that feedback, so its ledger adds no information. Public histories persist to allow arbitrary adaptive actions and a decision $\hat{e}_n$. No outside environment-specific data or hidden labels are provided.

Write $p_k = g(-.3 - .7k)$, $a_k = \Phi^{-1}(p_k)$, $v_k = d_k - c_P$, and $m(a) = |S(a)|$. Let $r_e(a)$ be expected net one-opportunity PnL. By (11),

$$r_0(a) = \sum_{s \in S(a)} \{v_{k_s} p_{k_s} - \sigma\theta\phi(a_{k_s})\} - \ell \mathbb{E}|F_+ - F_-|, \quad (15)$$

where an absent fill is set to zero. The selection sign is plus in r_1 . For two sides,

$$B_{k_b, k_a} = \int \phi(z) \Phi\left(\frac{a_{k_b} - \theta z}{\sqrt{1 - \theta^2}}\right) \Phi\left(\frac{a_{k_a} + \theta z}{\sqrt{1 - \theta^2}}\right) dz, \quad (16)$$

$$\mathbb{E}|F_+ - F_-| = p_{k_b} + p_{k_a} - 2B_{k_b, k_a}.$$

Thus the two-sided liquidation refund $2\ell B$ is included. Market buys/sells each have $r_e = -2\ell = -.054$, including entry and terminal spread/fee costs; abstention has $r_e = 0$.

Define

$$c_0 = \min_{k=0,1} \{\sigma\theta\phi(a_k) - v_k p_k\} > 0. \quad (17)$$

Positivity at this θ does not require simulation. Both $p_k \in (\Phi(-1), 1/2)$, hence $a_k \in (-1, 0)$. For example, the normal Mills bound gives $\Phi(-1) < \phi(1) < g(-1)$. Therefore

$$\sigma\theta\phi(a_k) - v_k p_k > .105\phi(1) - .049/2 > 0.$$

Equation (15), including nonnegative liquidation cost, proves $-r_0(a) \geq c_0 m(a)$ for **every** informative action. Abstention is uniquely optimal in $\mathcal{E}_0$; market orders also lose.

In $\mathcal{E}_1$, a single deep quote already has $r_1 = (.049 - .027)p_1 + .105\phi(a_1) > 0$; abstention is not optimal. The comparator π_e^* repeats an action in $\arg\max_a r_e(a)$, with deterministic tie breaking. This is attainable with the same actions and no future information. Enumeration of (15)–(16) selects both shallow quotes in $\mathcal{E}_1$, with expected PnL 0.0885245434. The proof requires only a different optimal decision and the analytic positive cost in $\mathcal{E}_0$, not a numerically delicate assertion about which profitable action is best.

4. Adaptive finite-sample evidence bound

For depth k , define

$$u_{0,k}(z) = \Phi\left(\frac{a_k - \theta z}{\sqrt{1 - \theta^2}}\right), \quad u_{1,k}(z) = u_{0,k}(-z), \quad I_k = \int \phi(z) \text{kl}\{u_{0,k}(z), u_{1,k}(z)\} dz, \quad (18)$$

with $\text{kl}(p, q) = p \log(p/q) + (1 - p) \log\{(1 - p)/(1 - q)\}$. Here I_k denotes information, distinct from quote distance d_k . Changing $z \mapsto -z$ shows that side and KL direction do not change I_k .

Let Q_e^a denote the actual selected-feedback law. Because Z is observed with the same law in both environments, conditional product additivity gives

$$D_a = \text{KL}(Q_0^a \| Q_1^a) = \sum_{s \in S(a)} I_{k_s}. \quad (19)$$

This does not assume unconditional fill independence. Market/abstention actions have $D_a = 0$.

Before thresholding, each selected latent execution variable has conditional Gaussian means $+\theta sz, -\theta sz$ and variance $1 - \theta^2$. Its divergence is $2\theta^2 z^2 / (1 - \theta^2)$. Thresholding data processing and $\mathbb{E}Z^2 = 1$ imply the explicit bound

$$I_k \leq D_G := \frac{2\theta^2}{1 - \theta^2}. \quad (20)$$

Allow any randomised history-dependent policy with the same decision rule in both environments and independent private randomness. Its action probability at a fixed public history is the same under either hypothesis and cancels in the transcript likelihood ratio. Multiplication of conditional outcome densities and the tower property give

$$\text{KL}(P_0^\pi \| P_1^\pi) = \mathbb{E}_0 \sum_{t=1}^n D_{A_t} = I_0 \mathbb{E}_0 N_0 + I_1 \mathbb{E}_0 N_1. \quad (21)$$

Here N_k counts **submitted sides** at depth k , including non-fills; $N = N_0 + N_1 \leq 2n$. It does not count successful fills. Deterministic transcript functions such as PnL add no divergence. All likelihood ratios exist and their KL is finite for $0 < \theta < 1$, by (20).

Suppose an environment decision satisfies

$$\max_{e=0,1} P_e^\pi(\hat{e}_n \neq e) \leq \delta, \quad 0 < \delta < 1/2. \quad (22)$$

The event $E = \{\hat{e}_n = 0\}$ has $P_0(E) \geq 1 - \delta$, $P_1(E) \leq \delta$. Data processing to its indicator and binary-KL monotonicity for $p > q$ imply

$$\text{KL}(P_0^\pi \| P_1^\pi) \geq \text{kl}(1 - \delta, \delta) = (1 - 2\delta) \log \frac{1 - \delta}{\delta} =: \beta_\delta. \quad (23)$$

Combining (19)–(23),

$$\boxed{\mathbb{E}_0 N \geq \frac{\beta_\delta}{I_{\max}} \geq \frac{\beta_\delta}{D_G}, \quad n \geq \frac{\beta_\delta}{2I_{\max}}, \quad I_{\max} = \max(I_0, I_1).} \quad (24)$$

Reverse KL symmetry gives the same side-count bound under $\mathcal{E}_1$. If M counts periods with at least one quote, $N \leq 2M$ also gives $\mathbb{E}_e M \geq \beta_\delta / (2I_{\max})$. Periods with only market orders or abstention add zero information. This distinguishes elapsed periods, quoted periods and submitted sides.

This change-of-measure mechanism is standard; see Lemma 1 of [Kaufmann, Cappé and Garivier \(2016\)](#). Equations (18)–(21) connect it to the actual feedback law. The fixed- n proof above is self-contained and invokes no asymptotic sample-complexity claim.

5. Economic lower bound

Define expected regret against the attainable comparator by

$$\mathcal{R}_e^\pi(n) = n \max_a r_e(a) - \mathbb{E}_e^\pi \sum_{t=1}^n Y_t. \quad (25)$$

Fresh innovations are independent of the adaptively chosen action conditional on the public past. The tower property, (17), and optimal abstention in $\mathcal{E}_0$ give

$$\mathcal{R}_0^\pi(n) = \mathbb{E}_0 \sum_t [-r_0(A_t)] \geq c_0 \mathbb{E}_0 N.$$

Uninformative market actions add positive regret. Therefore every strategy satisfying (22) pays

$$\boxed{\mathcal{R}_0^\pi(n) \geq \frac{c_0 \beta_\delta}{I_{\max}} \geq \frac{c_0(1-\theta^2)}{2\theta^2} (1-2\delta) \log \frac{1-\delta}{\delta}.} \quad (26)$$

Using full action costs, including liquidation, sharpens the constant:

$$\eta = \min_{a: D_a > 0} \frac{-r_0(a)}{D_a} > 0, \quad \boxed{\mathcal{R}_0^\pi(n) \geq \eta \beta_\delta.} \quad (27)$$

Every informative action satisfies $-r_0(a) \geq \eta D_a$; uninformative actions have nonnegative regret. Apply this to (21) and (23). This is the strongest constant from the single KL-to-cost comparison, not a claim of attainability. Its integral definitions are exact; their printed decimal evaluations are numerical.

Gaussian reward components are unbounded. The proof uses integrable conditional expectations and finite KL, not bounded-reward concentration. It establishes positive **expected** cost in the named environment; some realised exploratory trades can be profitable. An always-abstaining strategy avoids the cost by violating (22).

5.1 Numerical illustration at $\delta = .05$

Values are saved in `outputs/theory/constants.json` and `outputs/theory/actions.csv`:

Quantity	Value
Testing requirement $\beta_{.05}$	2.6499950812 nats
Gaussian upper bound per side D_G	0.2792022792
Exact one-side KL, shallow / deep	0.1674822241 / 0.1483585541
Minimum gross cost per side c_0	0.0214713573
Expected sides, Gaussian / exact-feedback bound	9.4913089 / 15.8225453
Necessary integer number of fixed opportunities	8
Regret bound, Gaussian / exact-side / exact-action	0.2037913 / 0.3397315 / 0.4846337
Cost per nat minimiser	Both deep quotes

Eight opportunities is a necessary lower bound, **not** a claim that eight suffice for 5% error. Expected side counts need not be integers. PnL units are the synthetic model’s monetary units.

Action class	r_0 , bad regime	r_1 , good regime	KL
Abstention	0	0	0
One shallow quote, either side	-0.0424342	0.0398809	0.1674822
One deep quote, either side	-0.0287328	0.0405662	0.1483586
Two shallow quotes	-0.0761056	0.0885245	0.3349644
One shallow and one deep, either orientation	-0.0658375	0.0857765	0.3158408
Two deep quotes	-0.0542639	0.0843340	0.2967171
Market buy or sell	-0.0540000	-0.0540000	0

These classes cover all eleven actions. Shallow quotes maximise information per side, but two deep quotes minimise bad-environment expected loss per nat. Maximising information and minimising its economic cost are different objectives.

6. A finite-horizon identification illustration

Repeatedly quote both deep sides and accumulate $L_t = \log(dP_0/dP_1)$ from selected feedback. Stop quoting when $|L_t| \geq B = \log(2/\delta)$, then abstain. At a fixed maximum horizon n , decide from the sign of the stopped likelihood ratio. This is an identification procedure, not the proposed trading policy.

For the always-quote action the Hellinger affinity is

$$\alpha = \int \sqrt{dQ_0^8 dQ_1^8} = \int \phi(z) \prod_{s=\pm 1} \left[\sqrt{c_{+,s} c_{-,s}} + \sqrt{(1 - c_{+,s})(1 - c_{-,s})} \right] dz < 1. \quad (28)$$

Under $\mathcal{E}_0$, the reciprocal likelihood ratio is a unit-mean martingale. At bounded stopping, Markov's inequality bounds an erroneous lower-boundary crossing by e^{-B} . If no boundary is reached, a wrong terminal decision is contained in $L_n \leq 0$ for an underlying always-quote experiment, whose probability is at most $\mathbb{E}_0 e^{-L_n/2} = \alpha^n$. The reciprocal argument under $\mathcal{E}_1$ gives

$$\max_e \Pr(\hat{e}_n \neq e) \leq e^{-B} + \alpha^n. \quad (29)$$

Potential later feedback is a distributional coupling used in the proof, not policy input. The policy only receives feedback from its actual quotes.

Numerical integration gives $\alpha = 0.9322425302$. At $n = 55$, $B = \log 40$, the analytic risk bound evaluates numerically to $0.04609066 < .05$. This numerical evaluation is subject to integration precision; it is not an interval certificate. Equation (29) with its exactly defined integral is non-asymptotic.

With seed 2026092404 and 40,000 independent repetitions per environment:

Environment	Error rate (Monte Carlo SE)	Mean submitted sides	Expected regret from quote counts
Bad $H = +1$	0.014625 (0.000600)	30.19065	0.819131
Good $H = -1$	0.013700 (0.000581)	30.00705	3.603542

Good-environment regret is high because this identification procedure abstains after stopping, even after detecting profitability. It is not an economic policy. Bad-environment loss exceeds the 0.484634 lower bound. Raw PnL, uncertainty and the adaptive KL compensator are in outputs/theory/identification.json. These experiments check implementation consistency and illustrate scale; they do not prove the theorem or its tightness.

7. Verification, attribution and scope

Run from the project root:

```
PYTHONPATH=src .venv/bin/python verification/verify_theory.py
```

The artifact outputs/theory/verification.json records hashes, seeds, runtime and tolerances. Independent full-feedback enumeration with 80- and 160-point Gaussian-Hermite quadrature agrees with the closed-moment/adaptive-integral calculation to below 5×10^{-16} . This agreement is evidence of stability, not certified interval integration.

An independent signed cash ledger using 250,000 raw normal triples per environment checks all eleven expected action rewards, shared-noise depth monotonicity and terminal liquidation. The largest discrepancy is 2.614 Monte Carlo standard errors, within the prespecified 6.5-SE simultaneous tolerance. These verification samples are separate from learner training and final policy evaluation.

Selected-feedback likelihoods, Gaussian conditioning, Bayesian filtering, Bellman recursions, KL chain rules, binary data processing and likelihood-ratio testing are established methods. The environment pair, all-action cost argument, constants and benchmark application are task-specific derivations. No originality is claimed for the underlying methods.

The theorem requires uniform identification accuracy in a fixed-sign reset experiment. It proves neither a regret bound for the switching controller nor an advantage from learning unknown parameters. Public signal, inventory, switching and horizon change the full problem; informative trades can be profitable. Matched empirical comparisons assess regime-information planning after the pilots. Stress losses combine changed opportunities and model mismatch, rather than isolate a causal cost of duration shape or filtering.

8. Pilot-aware statistical estimators

For environment e , independent pilot $j = 1, \dots, M$ and final episode $k = 1, \dots, n$, let D_{ejk} be the objective difference between two policies evaluated on the same exogenous tape. Here $M = 10, n = 100$. Define

$$\bar{D}_{ej} = n^{-1} \sum_k D_{ejk}, \quad \hat{\Delta}_e = M^{-1} \sum_j \bar{D}_{ej}, \quad \widehat{\text{SE}}_e^2 = S_e^2/M,$$

where S_e^2 is the sample variance of the ten pilot means. A descriptive two-sided interval is $\hat{\Delta}_e \pm t_{M-1, 1-\alpha/2} \widehat{\text{SE}}_e$. This treats pilot datasets as independently sampled units and retains the variability of fitting. It does not pretend that 300 sequential decisions are independent. If the pilot means are non-Gaussian, the Student interval is an empirical approximation, particularly with ten pilots.

To expose the two noise sources, let s_{ej}^2 be the within-pilot sample variance of paired episode differences. The estimated contribution of Monte Carlo noise to a pilot mean is $W_e = M^{-1} \sum_j s_{ej}^2/n$. Report the observed pilot-mean variance S_e^2 , W_e , and the method-of-moments excess $\max(S_e^2 - W_e, 0)$. Truncating that component at zero does **not** replace S_e^2 in the reported standard error.

For the fixed uniform mixture of nine environments,

$$\hat{\Delta} = \frac{1}{9} \sum_e \hat{\Delta}_e, \quad v_e = \widehat{\text{SE}}_e^2/81, \quad \widehat{\text{SE}}^2 = \sum_e v_e, \quad \nu = \frac{(\sum_e v_e)^2}{\sum_e v_e^2/(M-1)}.$$

The aggregate uses Welch–Satterthwaite Student quantiles with ν degrees of freedom. The environment grid is a fixed population with uniform weights, not nine random draws from all markets. Independently generated pilot/test streams across environments make the stratified variance calculation appropriate to this declared population. Policy differences are paired at episode level within each pilot; isolated marginal policy intervals cannot substitute for their paired difference.

The primary comparison was fixed as active versus matched noinfo on the nominal mixture. The acceptance threshold is 0.05 penalised-objective units per episode. Its one-sided 95% lower bound must exceed that threshold, and active net PnL must have a positive one-sided lower bound. This intersection does not select among many winners; it assesses one preselected candidate. Supplemental active-versus-six-comparator contrasts across nine environments and three variants form a family of 162. Their saved tables provide both descriptive 95% intervals and Bonferroni-adjusted intervals using $\alpha/162$. Adjustment addresses multiplicity under the interval approximation; it does not turn that approximation into an exact finite-sample guarantee.

Downside summaries are the empirical fifth PnL percentile, mean of observations at or below it, and loss frequency. They are descriptive tail estimates, not rare-event certificates. Gaussian innovations yield unbounded returns. No bounded-reward concentration result is used. Stress evidence is not used to choose a nominal policy, and nominal confidence says nothing distribution-free about arbitrary new regimes.

9. Practical-horizon numerical refinement

Both matched modes were checked independently at every public parameter point. Every remaining horizon 1 through 300 was checked. The table gives the last accepted belief-axis and integration-axis initial-state changes, maximized over horizons, with maximum common-state drift and worst per-horizon robust action disagreement across these two steps. The deployed belief grid is at least as fine as the accepted belief setting. All campaign tables use 161 integration nodes. For active (0.65, 0.002), the last integration check is 161 to 321 nodes at 1,281 beliefs; 321 nodes were a verification refinement and were not deployed. Other last integration checks compare 81 to 161 nodes. Earlier failures and all steps remain in `outputs/tables/all_refinement_steps.csv`. The limits, enforced at every horizon, were initial drift 0.01, common-state drift 0.03, and disagreement 0.005 among states with fine-grid action gap above 0.005. Numerical ties use the prespecified absolute 1e-12 rule; near-tie counts are separate. These checks are not certified error bounds.

θ	κ	Mode	Belief change	Integration change	Max drift	Robust mismatch
0.15	0.002	active	0.002093	0.000006	0.004054	0.0000
0.15	0.002	noinfo	0.000000	0.000000	0.003777	0.0000
0.15	0.02	active	0.000294	0.000001	0.000343	0.0000
0.15	0.02	noinfo	0.000000	0.000000	0.000189	0.0000
0.15	0.1	active	0.000068	0.000004	0.000099	0.0000
0.15	0.1	noinfo	0.000000	0.000000	0.000041	0.0000
0.35	0.002	active	0.005172	0.000047	0.008122	0.0000
0.35	0.002	noinfo	0.000000	0.000000	0.013212	0.0000
0.35	0.02	active	0.001389	0.000046	0.001605	0.0000
0.35	0.02	noinfo	0.000000	0.000000	0.000860	0.0000
0.35	0.1	active	0.000348	0.000077	0.000421	0.0000
0.35	0.1	noinfo	0.000000	0.000000	0.000135	0.0000
0.65	0.002	active	0.001425	0.000214	0.001939	0.0000
0.65	0.002	noinfo	0.000000	0.000000	0.006014	0.0000
0.65	0.02	active	0.003370	0.000169	0.003722	0.0000
0.65	0.02	noinfo	0.000000	0.000000	0.002391	0.0000
0.65	0.1	active	0.000736	0.000264	0.000885	0.0000
0.65	0.1	noinfo	0.000000	0.000000	0.000298	0.0000

10. Matched mechanism interventions

The original PDF asks to separate forecasting, inventory planning, current-regime inference and deliberate acquisition of future information. The initial report’s noinfo–myopic gap and local arguments did not identify all four policy effects. This supplementary study implements the missing comparisons in the prescribed reduced model: $T = 30$, $|q| \leq 2$, $\theta = .35$, $\kappa = .02$.

The [protocol](#) and [failure catalogue](#) were committed at ac7b70b before implementation or results. The separate seed 260924506 supplies 20,000 independent episodes shared by seven fixed policies: 140,000 policy records, with no fitting, tuning or pilot uncertainty. This new experiment does not replace or pool with the original 189,000-record finite-pilot campaign.

Four matched comparisons and results

All rules act in the **same true simulator**, with the same cash accounting, inventory constraints, fees, horizon, terminal liquidation and observable public signal. The four effects below are paired differences in penalised objective per episode. Brackets give approximate **simultaneous 95% Student/Bonferroni intervals for the four prespecified contrasts**, with 19,999 degrees of freedom.

Channel	Changed decision input or operation	Effect and simultaneous interval
Predictive drift use	Filtered noinfo minus no-forecast; remove only $.03xE[q']$ from every planning reward, keeping signal-dependent fills, transitions and correct filtering.	+0.190707 [0.156094, 0.225320]
Current regime inference use	Filtered noinfo minus blind-regime; identical noinfo tables, but the blind rule routes belief $.5$ into every decision. It keeps nonzero true dependence and does not use its shadow filter.	+0.241601 [0.214178, 0.269023]
Action-induced inventory continuation	Filtered noinfo minus inventory-off; replace only q' by current q in the same continuation before the actual last step. Keep the last-step scores exactly equal.	+0.085447 [0.056881, 0.114013]
Future information valuation	Active minus filtered noinfo; only feedback conditioning in future planning changes. Both update current beliefs from their own selected observations.	+0.026197 [0.009232, 0.043161]

Each interval is above zero on this fixed synthetic configuration. This supplies four distinct operational policy effects. **They are not additive components of PnL**, and they do not estimate a universal premium for each mechanism. Policy actions subsequently change which evidence and inventories are encountered. The first three comparisons are anchored at the same filtered noinfo policy; only the fourth changes future-feedback valuation.

Exact changes to the decision rules

Let r be the expected current marked-wealth increment less the pre-decision risk charge, $G(q) = -(h + c_T)|q|$, and $\psi(b) = \kappa + (1 - 2\kappa)b$. The noinfo planning recursion is

$$Q_m^N(q, x, b, a) = r(q, x, b, a) + E_b[V_{m-1}^N(q', X', \psi(b))], \quad V_0^N = G.$$

The runtime noinfo policy still filters its realised return/fill feedback. Its stored no-learning planning value therefore differs from its realised objective.

For **forecast use**, solve this same recursion with $r^{F0} = r - \mu x E[q' \mid q, x, a]$. A passive action has $E[q'] = q + \sum_{s \in S(a)} sp_{s, k_s}(x)$; a market order has $E[q'] = q + s$. The actual simulator and filter retain $\mu = .03$; in particular, $z = (R - .03x)/.30$ remains the observed innovation. The policy still uses X for fill probabilities and future public transitions. This comparison measures the use of the specified predictive **return drift**, not the value of hiding the entire signal or fitting an unknown forecasting model.

For **current inference**, the enabled rule selects using $Q_m^N(q, x, b_t, a)$ and the blind rule uses $Q_m^N(q, x, .5, a)$. The symmetric regime begins at its stationary prior, so $\psi(.5) = .5$. The blind rule’s shadow posterior is saved for verification but never routed into decisions. This comparison includes the economic consequences of using evidence the policy has actually acquired. It does not assume information was free, or force two diverging policies to have identical histories. It does not change the execution model to $\theta = 0$, as the older independence baseline does.

For **inventory continuation**, use exactly the same V^N and define

$$Q_m^{I0}(q, x, b, a) = \begin{cases} Q_1^N(q, x, b, a), & m = 1, \\ r(q, x, b, a) + E[V_{m-1}^N(q, X', \psi(b))], & m \geq 2. \end{cases}$$

Before the last step, this deletes precisely

$$D_m(q, x, b, a) = E[V_{m-1}^N(q', X', \psi(b)) - V_{m-1}^N(q, X', \psi(b))].$$

Neither K nor $\psi(b)$ depends on the action. Consequently, all action-dependent future continuation in noinfo is mediated by the change from q to q' . The marginal selected-fill vector is also sign-invariant: substituting $z \mapsto -z$ exchanges the two regime likelihoods under the symmetric normal density. This includes joint two-sided fills.

The ablated score intentionally omits the effect of today’s action on inventory supplied to continuation. It is **not** the optimum of an alternative physical market. The simulator still changes inventory, enforces the same safe actions and charges the same current risk; the rule observes the real inventory again next period. At the last decision it retains $E[G(q')]$ exactly, with one real terminal liquidation. Before then its action-independent continuation means it maximises immediate reward. This differs from the existing myopic rule, which includes a hypothetical immediate liquidation at every decision.

Future signal opportunities that depend on inherited inventory remain part of D_m . The measured effect is **multi-period use of action-induced inventory consequences**, including interactions with forecasts, constraints and future executions. It is not a pure contribution of λq^2 , nor a change in the public-signal process. This exact common-continuation construction removes the earlier ambiguity of a generic noinfo-myopic comparison.

For **future information**, active replaces $\psi(b)$ in the planning continuation by $\psi(B)$, where B is the feedback posterior. The original controlled witness remains: at $(m, q, x, b) = (30, 0, 0, .75)$, both deep quotes have immediate expected reward $-.00829323$; suppressing current feedback valuation with the very same active continuation instead selects abstention, with an active-Q gap $.00349678$. This is a state illustration selected after solving, not an extra statistical test.

Policy outcomes and validation

Policy	Mean objective	Monte Carlo SE	Mean net PnL
Active	0.990439	0.016146	1.071387
Filtered noinfo	0.964242	0.016480	1.052758
No forecast	0.773536	0.013283	0.813878
Blind regime	0.722642	0.016060	0.788921
Inventory off	0.878795	0.018261	0.993802
Myopic with immediate liquidation	0.839049	0.013645	0.886818

Policy	Mean objective	Monte Carlo SE	Mean net PnL
Current-regime full-information reference	1.459878	0.015778	1.544889

The active, blind-regime and full-information means agree with their correctly matched planning values within 0.77 Monte Carlo SE. No such calibration claim is made for filtering noinfo, no-forecast or inventory-off planning scores. The [raw summary](#) also reports ordinary 95% intervals, fees, liquidation, inventory exposure and all eleven action frequencies. Raw episode rows and complete selected-action arrays allow every comparison and ledger to be checked.

Cold numerical builds use 321 beliefs and 161 integration nodes. Separate 161-to-321 belief and 81-to-161 quadrature comparisons pass the unchanged reduced thresholds at every horizon for active, noinfo, no-forecast and inventory-off scores. These are empirical stability checks, not certified bounds. Independent fill enumeration checks 1,845 inventory-deletion cases with maximum error 8.2×10^{-16} ; the drift identity error is below 2.0×10^{-17} . Exact last-step equality, a nonconstant synthetic continuation and zero effect for inventory-independent continuation also pass. The comparisons cover all 144,450 grid states/horizons; aggregate disagreement counts, maximum score losses and selected illustrations are retained.

Two fresh processes sharing no control cache reproduce all saved episode, action, trace, summary and numerical-check bytes and all table fingerprints on the recorded stack. The standalone independent verifier imports no production simulator, filter or statistical code: it regenerates paired tapes, replays all 4.2 million selected decisions from the cash-flow equations, reconstructs the 2,520 saved public decision traces and filters, and recalculates all four intervals. Its corruption probes reject incomplete, stale or altered inputs. These are implementation-team checks, not a new external reviewer sign-off.

Earlier evidence retained separately

The canonical nine-environment active-minus-noinfo primary remains +1.3925613223, pilot-aware approximate 95% interval [1.2953383014, 1.4897843432]. Its corrected replay uses the original streams and adds no independent sample. The new reduced study uses a separate root seed and known parameters; it cannot expand the canonical deployment population or establish persistent parameter learning.

The earlier nominal noinfo-minus-myopic contrast remains **post hoc and descriptive**: +2.4774821599 [2.2825822518, 2.6723820680] across 90 fitted pilots. It changes continuation and internal liquidation timing together and is not the inventory treatment defined above. Its specification, results and independent arithmetic remain in Git history and the `mechanism_contrasts.json` artifact alongside the new study; it is outside both prespecified contrast families. Local drift sensitivities and the hypothetical free-signal Jensen argument remain valid analytical context, not replacements for the implemented interventions.